

Lecture 9

Theory of Square Linear System II: LU Decomposition

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Recall

In the last lecture ...

- Definition of the inverse of a matrix.
- Computed matrix inverse for simple matrices.
- Important properties of matrix inverse.

Today's Lecture: Outline

Today ... LU and PLU decomposition

1. Elementary Matrix
2. LU Decomposition

After this lecture, you should be able to

1. Compute the elementary matrix for each operation
2. Compute LU decomposition of simple matrices

Strang's book: Sec 2.5, 2.6

Part I Elementary Matrix

Strang's book v5 Sec 2.6

[~35 mins]

Recall: Elementary Row Operations In Lec 4

Definition (Elementary Row Operations) (初等行变换)

1. [Multiplication] Multiply a row by a **nonzero** scalar:

$$R_i \rightarrow \alpha R_i, \quad \alpha \neq 0$$

2. [Adding a Scaled Row] Add a scaled multiple of one row to another row:

$$R_i \rightarrow R_i + \beta R_j$$

3. [Interchange] Swap the positions of two rows:

$$R_i \leftrightarrow R_j$$

→ finally this

→ then this

↑ order today

→ start with this one

Row Operation = Matrix Multiplication?

$$M_1 = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \quad \text{after interchange } R_1 \text{ and } R_3: \quad M_2 = \begin{bmatrix} e & f \\ c & d \\ a & b \end{bmatrix}$$

Can we find a matrix $A \in \mathbb{R}^{3 \times 3}$ such that $M_2 = AM_1$ for all $a, b, c, d, e, f, g \in \mathbb{R}$?

More general question:

Suppose $M_1 \in \mathbb{R}^{3 \times k}$, and $M_1 \xrightarrow{R_1 \leftrightarrow R_3} M_2$.

Can we find a matrix $A \in \mathbb{R}^{3 \times 3}$ such that $M_2 = AM_1$ for any M_1 ?

Answer: yes, $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

How to Derive E_{R_i, R_j} from scratch?

Assume “swapping rows” = “matrix multiplication”.
Then what will the matrix be like?

$$M_1 \xrightarrow{R_1 \leftrightarrow R_3} M_2$$

Step 1: Looking for a candidate.

$$\Leftrightarrow AM_1 = M_2, \forall M_1$$

Specify M_1 to be I_3 .

Trick: Apply the operation to identity matrix.

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

We look for $A \in \mathbb{R}^{3 \times 3}$ s.t.: $\underline{A}I_n =$

find unknown
matrix A

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

How to Derive E_{R_i, R_j} from scratch?

Assume “swapping rows” = “matrix multiplication”.
Then what will the matrix be like?

Step 1: Looking for a candidate.

Trick: Apply the operation to **identity matrix**.

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

We look for $A \in \mathbb{R}^{3 \times 3}$ s.t.: $AI_n = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

Compute the candidate as follows:

$A \cdot I_3 = A$.

$$\textcircled{A} = A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \Rightarrow A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Step 2: Verify that multiplying M by this A indeed swaps rows of M.
Next page.

Verify: Row Operation = Matrix Multiplication

Step 2: **Verify that** multiplying M by this A indeed swaps rows of M .

For simplicity, we only consider a 3 by 2 matrix M_1 .

(a complete verification requires checking all 3 by k matrices, which we skip here)

Task.

translate

$$M_1 = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \quad \text{after interchange } R_1 \text{ and } R_3: \quad M_2 = \begin{bmatrix} e & f \\ c & d \\ a & b \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Need to verify $A \cdot M_1 = M_2$.

Verify: Row Operation = Matrix Multiplication

Step 2: **Verify that** multiplying M by this A indeed swaps rows of M .

For simplicity, we only consider a 3 by 2 matrix M_1 .

(a complete verification requires checking all 3 by k matrices, which we skip here)

Task: $M_1 = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$ after interchange R_1 and R_3 : $M_2 = \begin{bmatrix} e & f \\ c & d \\ a & b \end{bmatrix}$

$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

Need to verify $AM_1 = M_2$.

$[0, 0, 1] \begin{bmatrix} a \\ c \\ e \end{bmatrix} = e$

Verify below:

$$AM_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} = \begin{bmatrix} e & f \\ c & d \\ a & b \end{bmatrix} = M_2$$

The Process of Finding Corresponding Matrix

Goal: Find A such that AM (where M is $n \times k$)

$\iff$ applying $R_i \leftrightarrow R_j$ to M . $(*)$

Q1: How to compute such a matrix A ?

Answer: Apply $R_i \leftrightarrow R_j$ to I_n to get a matrix

$$A \triangleq E_{R_i, R_j}.$$

This is just a definition. We need to **verify** it satisfies the desired property.

Q2: Does the matrix achieve the desired goal $(*)$?

need to verify: Let $M \xrightarrow{R_i \leftrightarrow R_j} M'$, then $E_{R_i, R_j} M = M'$

[skip the step here]

Exercise: Compute Matrix w.r.t. Row Exchange

Suppose $M_1 \in \mathbb{R}^{4 \times k}$, and $M_1 \xrightarrow{R_2 \leftrightarrow R_4} M_2$.

Can we find a matrix $A \in \mathbb{R}^{4 \times 4}$ such that $M_2 = AM_1$ for any M_1 ?

Procedure of Computation:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_4} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \triangleq A.$$

This matrix A is the desired matrix;
left-multiplying a matrix M with 4 rows by A corresponds to
swapping R_2 and R_4 of M .

[skip the proof here]

Operation 1: Swapping Rows

Swap Two Rows ($R_i \leftrightarrow R_j$)

Definition

The **row exchange matrix** $E_{R_i, R_j} \in \mathbb{R}^{n \times n}$ is the matrix obtained by swapping the i -th and j -th rows of the $n \times n$ identity matrix I_n .

Expression of the matrix

$$E_{R_i, R_j} = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 & & \\ & & & & & \ddots & \\ & & & & & & 1 & & \\ & & & & & & & \ddots & \\ & & & & & & & & 1 & \\ & & & & & & & & & \ddots \\ & & & & & & & & & & 1 \end{bmatrix}$$

Handwritten annotations:

- Red circles around the entries at (i, i) , (i, j) , (j, i) , and (j, j) .
- Red arrows pointing from the (i, i) and (j, j) positions to the (j, i) and (i, j) positions, respectively.
- Blue circles around the entries at (i, i) and (j, j) .
- Blue circles around the entries at (j, i) and (i, j) .
- Blue labels: (i, i) , (i, j) , (j, i) , (j, j) .
- Blue label: j -th row (pointing to the j -th row).
- Blue label: i -th column (pointing to the i -th column).
- Blue label: j -th column (pointing to the j -th column).
- Blue label: i -th row (pointing to the i -th row).
- Blue label: j -th row (pointing to the j -th row).

Structure of the matrix

- The diagonal entries are 1, except for the (i, i) and (j, j) positions, which are 0.
- The (i, j) and (j, i) entries are 1.
- All other entries are 0.

Operation 1: Swapping Rows

Property

For any matrix $A \in \mathbb{R}^{n \times k}$ (where k is any positive integer),
swapping the i -th and j -th rows of A results in $E_{R_i, R_j} A$.

Example

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} w & x \\ y & z \end{bmatrix} = \begin{bmatrix} y & z \\ w & x \end{bmatrix}$$
$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} = \begin{bmatrix} a_{31} & a_{32} \\ a_{21} & a_{22} \\ a_{11} & a_{12} \end{bmatrix}$$

What about other row operations?

Assume “elementary row operations” can be achieved by matrix multiplication.

Then what will they be like?

Trick: Apply it to **identity matrix**

Add a Scaled Row: Add βR_2 to R_4 (in a 4×4 matrix)

$$\begin{array}{c} R_2 \\ R_4 \end{array} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\beta R_2 + R_4} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & \beta & 0 & 1 \end{bmatrix} \quad E_{\beta R_2 + R_4}$$

Scaling rows: Scale R_2 by α (in a 4×4 matrix)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\alpha R_2} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \alpha & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad E_{\alpha R_2}$$

Operation 2: Add a Scaled Row

Add a scaled row

Definition

The matrix $E_{\beta R_i + R_j} \in \mathbb{R}^{n \times n}$ is obtained by performing the row operation $R_j \rightarrow \beta R_i + R_j$ on the $n \times n$ identity matrix I_n .

Expression of the matrix

$$E_{\beta R_i + R_j} = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \\ & & \beta & & \\ & & & 1 & \\ & & & & \ddots \\ & & & & & 1 \end{bmatrix}$$

ith row

jth row

ith column *jth column*

(j,i)

Structure of the matrix

- The diagonal entries are all 1.
- The (j, i) -th entry (the entry in the j -th row and i -th column) is β .
- All other entries are 0.

Operation 2: Add a Scaled Row

Property (Add a Scaled Row to Another)

Suppose $E_{\beta R_i + R_j} \in \mathbb{R}^{m \times m}$, the result of $E_{\beta R_i + R_j} A (\alpha \neq 0)$ is to multiply each element of i th row of matrix A by α , then add them into the j th row while keeping i th row unchanged.

Example

$$E_{2R_1 + R_2} \begin{bmatrix} w & x \\ y & z \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} w & x \\ y & z \end{bmatrix} = \begin{bmatrix} w & x \\ 2w+y & 2x+z \end{bmatrix}$$

$$E_{-2R_1 + R_3} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \\ = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ -2a_{11} + a_{31} & -2a_{12} + a_{32} \end{bmatrix}$$

Operation 3: Multiply a Row by a Nonzero

Multiply a Row by a Nonzero Scalar

Definition

The matrix $E_{\alpha R_i} \in \mathbb{R}^{n \times n}$ is obtained by performing the row operation $R_i \rightarrow \alpha R_i (\alpha \neq 0)$ on the $n \times n$ identity matrix I_n .

Expression of the matrix

$$E_{\alpha R_i} = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & & \alpha & \\ & & & & 1 \\ & & & & & \ddots \\ & & & & & & 1 \end{bmatrix}$$

ith column

ith row

Structure of the matrix

- The diagonal entries are all 1, except for the (i, i) -th entry, which is α .
- All other entries are 0.

Operation 3: Multiply a Row by a Nonzero

Property (Multiply a Row by a Nonzero Scalar)

Suppose $E_{\alpha R_i} \in \mathbb{R}^{m \times m} (\alpha \neq 0)$, the result of $E_{\alpha R_i} A$ is just to multiply each element of i th row of matrix A by α .

Example

$$\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} w & x \\ y & z \end{bmatrix} = \begin{bmatrix} w & x \\ 3y & 3z \end{bmatrix}$$
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ 2a_{31} & 2a_{32} \end{bmatrix}$$

Remark: Diagonal matrix times a matrix is just multiplying each row by a diagonal entry

$$\text{e.g. } \begin{bmatrix} \alpha_1 & 0 & 0 \\ 0 & \alpha_2 & 0 \\ 0 & 0 & \alpha_3 \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \\ R_3 \end{bmatrix} = \begin{bmatrix} \alpha_1 R_1 \\ \alpha_2 R_2 \\ \alpha_3 R_3 \end{bmatrix}$$

↙
partitioned matrix product.

Elementary Matrix

Elementary Matrix:

The matrices corresponding to a single **elementary row operation** are called **elementary matrices**.

Elementary matrices include three types of matrices:

- E_{R_i, R_j} (row exchange)
- $E_{\beta R_i + R_j}$ (adding a scaled row)
- $E_{\alpha R_i}$ (row scaling)

For a given matrix A , performing elementary row operation for A is equivalent to premultiplying A by the corresponding elementary matrix.

Elementary Matrix

Exercise Are the following matrices elementary matrices?

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

✓

$$\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

✓

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$$

✓

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

✗

What are the Shapes? Triangular, diagonal or none?

Elementary matrix	Diagonal	Upper triangular	Lower triangular
1. $E_{R_1, R_3} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$			
2. $E_{3R_2} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$			
3. $E_{3R_1 + R_3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$			
4. $E_{3R_3 + R_1} = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$			



Note: If $i < j$, add βR_i to R_j (eliminate down, Phase 1 of GJ-E): lower triangular

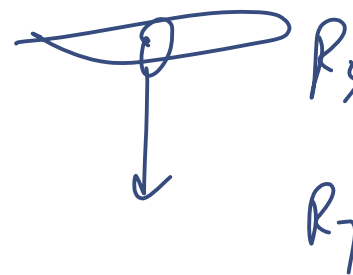
If $i > j$, add βR_i to R_j (eliminate up, Phase 2 of GJ-E): upper triangular

Exercise: Upper or Lower Triangular?

For each of the following elementary matrices, determine whether it is an **upper triangular** or **lower triangular** matrix:

1. $E_{2R_3+R_7}$

- A) Upper triangular
- B) Lower triangular ✓



2. $E_{3R_{100}+R_{57}}$

- A) Upper triangular ✓
- B) Lower triangular

What Is the Inverse of an Elementary Matrix?

Example

$$R_1 \leftrightarrow R_2 \quad \overbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}^{\text{elementary matrix}}^{-1} = \overbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}^{\text{elementary matrix}} \rightarrow R_1 \leftrightarrow R_2$$

$$3R_2 \quad \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \quad \frac{1}{3} R_2$$

$$3R_1 + R_3 \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \quad -3R_1 + R_3$$

Question: What Is the Inverse of an Elementary Matrix?

Answer: Matrix corresponding to the reverse operation.

Reverse Operations: Examples

$$A = \begin{bmatrix} \vec{a} \\ \vec{b} \\ \vec{c} \end{bmatrix} \xrightarrow{R_3 \rightarrow 3R_1 + R_3} \begin{bmatrix} \vec{a} \\ \vec{b} \\ 3\vec{a} + \vec{c} \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - 3R_1} \begin{bmatrix} \vec{a} \\ \vec{b} \\ \vec{c} \end{bmatrix} = A$$

Apply two operations $\iff$ Doing nothing

They are reverse operations.

Examples of real-world reverse operations.

- Turning a light on and **off**.
- Wearing and taking off shoes.
- **Sleeping** and waking up.

Non-Reversible Operations:

- Breaking a light bulb.
- Burning shoes.

Inverse of Elementary Matrix

Proposition (Inverse of Elementary Matrices)

$$E_{R_i R_j}^{-1} = E_{R_i R_j}, \quad \text{corresponding to the reverse row operation: } R_i \leftrightarrow R_j$$

$$E_{\alpha R_i}^{-1} = E_{\frac{1}{\alpha} R_i} \quad (\alpha \neq 0), \quad \text{corresponding to the reverse row operation: } R_i \rightarrow \frac{1}{\alpha} R_i$$

$$E_{\beta R_i + R_j}^{-1} = E_{-\beta R_i + R_j}, \quad \text{corresponding to the reverse row operation: } R_j \rightarrow -\beta R_i + R_j$$

Remark. The inverse of the elementary matrices corresponding to the reverse row operations belongs to the same type of elementary matrices.

What are the shapes of these inverses? Recall:

Elementary matrix	Diagonal	Upper triangular	Lower triangular
1. $E_{R_1, R_3} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$	×	×	×
2. $E_{3R_2} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	✓	✓	✓
3. $E_{3R_1 + R_3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$	×	×	✓
4. $E_{3R_3 + R_1} = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	×	✓	×

What are the Shapes of Inverse of them?

Type I:
Not triangular

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Type I inverse:
Not triangular

Type II:
Diagonal

$$\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{3} \end{bmatrix}$$

Type II inverse:
diagonal

Type III:
Lower triangular
Or
Upper triangular

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$

Type III inverse:
lower triangular
Or
upper triangular.

Same shape as the original matrix.

Recall: If $i < j$, add βR_i to R_j (eliminate down, Phase 1 of GJ-E): lower triangular

If $i > j$, add βR_i to R_j (eliminate up, Phase 2 of GJ-E): upper triangular

Exercise: Judge T or F

J1 Any elementary matrix is a square matrix.

T

J2 Any elementary matrix is invertible.

T

J3 Any elementary matrix is a triangular matrix.

F

row exchange matrix is not triangular.

Part II Computing LU Decomposition

Part II.1 GJ-E Sequence

Strang's book Sec 2.6

[~10 mins]

“Row Operation $\Leftrightarrow$ Matrix Multiplication” in GJ-E

We have learned two things:

(S1) (A single) elementary row operation
Is (a single) matrix multiplication. [last section]

(S2) GJ-E is a bunch of elementary row operations.
[Lec 04,07]

(Corollary)

(S3) GJ-E is a bunch of matrix multiplications.

3 by 3 Example

Coefficient matrix

$$A = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 5 & 2 \\ 4 & -1 & 9 \end{bmatrix}$$

One step of elimination:

$$\begin{bmatrix} 2 & 4 & 2 \\ 1 & 5 & 2 \\ 4 & -1 & 9 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - \frac{1}{2}R_1} \begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 1 \\ 4 & -1 & 9 \end{bmatrix}$$

Express as matrix multiplication:

Denote $E_1 = E_{-\frac{1}{2}R_1+R_2} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Then $E_1 A = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & 2 \\ 1 & 5 & 2 \\ 4 & -1 & 9 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 1 \\ 4 & -1 & 9 \end{bmatrix}$

3 by 3 Example

Coefficient matrix

$$A = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 5 & 2 \\ 4 & -1 & 9 \end{bmatrix}$$

Gaussian elimination:

$$\begin{bmatrix} 2 & 4 & 2 \\ 1 & 5 & 2 \\ 4 & -1 & 9 \end{bmatrix} \xrightarrow[\substack{R_2 \rightarrow -\frac{1}{2}R_1 + R_2 \\ R_3 \rightarrow -2R_1 + R_3}]{\substack{R_2 \rightarrow -\frac{1}{2}R_1 + R_2 \\ R_3 \rightarrow -2R_1 + R_3}} \begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 1 \\ 0 & -9 & 5 \end{bmatrix} \xrightarrow{R_3 \rightarrow -(-3)R_2 + R_3} \begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 1 \\ 0 & 0 & 8 \end{bmatrix} \triangleq U$$

Express as matrix multiplication:

$$E_2 E_1 A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 1 \\ 0 & -9 & 5 \end{bmatrix}$$

$$E_3 (E_2 E_1 A) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 1 \\ 0 & -9 & 5 \end{bmatrix} = U$$

3 by 3 Example: Expression of A

$$A = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 5 & 2 \\ 4 & -1 & 9 \end{bmatrix} \quad E_3(E_2E_1A) = U.$$

$$E_3E_2E_1 A = U$$

Matrix expression of Phase I of GJ-E.

where each E_i is an elementary row matrix,
and U is an upper triangular matrix.

This is a nice result in math.

Part II Computing LU Decomposition

Part II.2 LU Decomposition

Strang's book Sec 2.6

[~25 mins]

Expression of A: So What?

$$A = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 5 & 2 \\ 4 & -1 & 9 \end{bmatrix} \quad E_3(E_2E_1A) = U.$$

$$E_3E_2E_1 A = U$$

Matrix expression of Phase I of GJ-E [good case]
where each E_i is an elementary row matrix,
and U is an upper triangular matrix.

This is a nice result in math.

Any benefit?

Next: we get an interesting expression of A.

3 by 3 Example: Expression of A

Denote $M = E_3 E_2 E_1$, then $MA = U$.

From $MA = U$, can we get an expression of A?

Define $L = M^{-1}$ (if M is invertible), then

$$A = LU.$$



Here, L is a lower triangular matrix (proved next), and
U is an upper triangular matrix (by the process of GJ-E)

Claim 1 (informal):

Under certain condition, L is a lower triangular matrix.

日期: /

$$MA = U \Rightarrow A = ?$$

$$(B^{-1}) \quad B \cdot \vec{x} = \overset{(B^{-1})}{\vec{u}} \Rightarrow \vec{x} = B^{-1} \vec{u} \quad (\text{if } B^{-1} \exists)$$

$$(B^{-1}) \quad \underline{B} \cdot \underline{X} = \overset{(B^{-1})}{U} \Rightarrow X = B^{-1} U \quad (\text{if } B^{-1} \exists)$$

change symbols: change B to M ; change X to A .

$$MA = U \Rightarrow A = M^{-1} U \quad (\text{if } M^{-1} \exists)$$

3 by 3 Example: Verify L is Lower Triangular

Assumption 9.1. There is no row-exchange in the Gauss-Jordan process for A.

Denote $M = E_3 E_2 E_1$. $L = M^{-1}$ exists and $L = E_1^{-1} E_2^{-1} E_3^{-1}$

Handwritten notes: "invertible" above E_3 ; "since product of invertible matrices is invertible (Lec 8)" above $L = M^{-1}$

Claim 1: L is a lower triangular matrix.

Proof:

Step 1: E_1, E_2, E_2 are lower triangular matrices. (1)

i) No row exchange: so they are Type II or III, so are triangular.

ii) E_1, E_2, E_3 correspond to the forward phase, so lower triangular.

Step 2: $E_3^{-1}, E_2^{-1}, E_1^{-1}$ are lower triangular matrices. (2)

By Property 7.1 on inverse of triangular matrices, in Lec 7.

Step 3: Product of lower triangular matrices is lower triangular. (3)

Proved in homework 2 (though proved for upper triangular matrices).

Combining (2) and (3): $L = E_1^{-1} E_2^{-1} E_3^{-1}$ is a lower triangular matrix.

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Essence of the proof:

$$\begin{array}{ccccc} E_i & \longrightarrow & E_i^{-1} & \longrightarrow & E_i^{-1} E_i^{-1} E_i^{-1} \\ \text{(elementary)} & & \text{(inverse)} & & \text{(product)} \end{array}$$

properties on shape of them (lower triangular or not).

n by n **No-Row-Exchange Case**: LU Decomposition

Coefficient matrix **A**; square matrix.

(S1) Gauss-Jordan elimination (forward part; assume no row exchange)

$$A \rightarrow A_1 \rightarrow A_2 \dots \rightarrow U$$

U is upper triangular.

(S2) Express GE as matrix multiplication:

$$\bar{E}_k \bar{E}_{k-1} \dots \bar{E}_1 A = U.$$

Express matrix A as matrix product:

$$A = \bar{E}_1^{-1} \bar{E}_2^{-1} \dots \bar{E}_k^{-1} U.$$

(S3) Denote $L = \bar{E}_1^{-1} \dots \bar{E}_k^{-1}$ it is lower triangular.

since only involves Type-I (eliminating down) and Type-II elementary matrices.

Then

(S4)
$$A = LU.$$

How to Compute LU Decomposition?

Previous page:

explains the major steps to **derive** LU decomposition.

[For 1st time learners to understand LU;
and for conceptual problems in hw/exam]

Next: we present the steps to **compute LU decomposition.**
(for computation problems in hw/exam)

Computing LU Decomposition

Algorithm 9.1 (compute LU decomposition)

Step 1: Forward elimination.

Run forward elimination, till get upper triangular matrix U.

IF row exchange needed:

STOP and report: the algorithm fails to generate LU decomposition.

ELSE (no row exchange) Go to Step 2.

Step 2 Record elementary matrices.

Record elementary matrices $E_1, \dots, E_k$ in Step 1.

Step 3: Compute L.

Compute $L = E_1^{-1} E_2^{-1} \dots E_k^{-1}$.

Remark:

Can use row operations to compute L.

Step 4: Conclusion.

A has an LU decomposition $A = LU$,

where L = blabla (obtained in Step 3)

U = blabla (recorded in Step 1)

3 by 3 Example: Compute LU

Step 1 Gauss-Jordan elimination (only the forward elimination part)

$$\begin{bmatrix} 2 & 4 & 2 \\ 1 & 5 & 2 \\ 4 & -1 & 9 \end{bmatrix} \xrightarrow[\substack{R_2 \rightarrow -\frac{1}{2}R_1 + R_2 \\ R_3 \rightarrow -2R_1 + R_3}]{\substack{R_2 \rightarrow -\frac{1}{2}R_1 + R_2 \\ R_3 \rightarrow -2R_1 + R_3}} \begin{bmatrix} 2 & 4 & 2 \\ 0 & \boxed{3} & 1 \\ 0 & -9 & 5 \end{bmatrix} \xrightarrow{R_3 \rightarrow -(-3)R_2 + R_3} \begin{bmatrix} 2 & 4 & 2 \\ 0 & \boxed{3} & 1 \\ 0 & 0 & 8 \end{bmatrix} \triangleq U.$$

Check: No row exchange in the process; continue.

Step 2: Record elementary matrices.

$$E_1, E_2, E_3 = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$$

3 by 3 Example: Compute LU

Step 1 Gauss-Jordan elimination (only the forward elimination part)

$$\begin{bmatrix} 2 & 4 & 2 \\ 1 & 5 & 2 \\ 4 & -1 & 9 \end{bmatrix} \xrightarrow{\substack{R_2 \rightarrow -\frac{1}{2}R_1 + R_2 \\ R_3 \rightarrow -2R_1 + R_3}} \begin{bmatrix} 2 & 4 & 2 \\ 0 & \boxed{3} & 1 \\ 0 & -9 & 5 \end{bmatrix} \xrightarrow{R_3 \rightarrow -(-3)R_2 + R_3} \begin{bmatrix} 2 & 4 & 2 \\ 0 & \boxed{3} & 1 \\ 0 & 0 & 8 \end{bmatrix} \triangleq U.$$

Check: No row exchange in the process; continue.

Step 2: Record elementary matrices.

$$E_1, E_2, E_3 = \begin{bmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$$

Step 3: Compute L by (10.3).

$$L = E_1^{-1}E_2^{-1}E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}^{-1}$$

$\frac{1}{2} = \text{opposite number of } -1/2.$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 2 & -3 & 1 \end{bmatrix}$$

Step 4: Conclusion.

A has an LU decomposition $A = LU$, where $L = \text{blabla}$, $U = \text{blabla}$.

[Next page]

Step 3: Compute L

product of elementary matrices

Compute $L = E_1^{-1} E_2^{-1} E_3^{-1}$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} = ?$$

Trick: Matrix multiplication can be done by row operations!

first applying the row operation corresponding to E_2^{-1} ($R_3 \rightarrow R_3 + 2R_1$),

then applying the row operation corresponding to E_1^{-1} ($R_2 \rightarrow R_2 + \frac{1}{2}R_1$).

Row Operation Sequence

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + 2R_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -3 & 1 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 + \frac{1}{2}R_1} \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 2 & -3 & 1 \end{bmatrix}$$

E_3^{-1} E_2^{-1} multiply E_1^{-1} multiply

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If you're not sure about your solution. check product.

$$A = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 5 & 2 \\ 4 & -1 & 9 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 2 & -3 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 1 \\ 0 & 0 & 8 \end{bmatrix}$$

Verify $A = LU$.

$$LU = \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 2 & -3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 1 \\ 0 & 0 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 4 & 2 \\ 1 & 5 & 2 \\ 4 & -1 & 9 \end{bmatrix} \text{ is indeed } A.$$

Exercise: Computing LU Decomposition

Step 1 Gauss-Jordan elimination (only the forward elimination part)

$$\begin{bmatrix} 2 & 4 & 6 \\ 2 & 5 & 3 \\ 4 & -1 & 2 \end{bmatrix} \xrightarrow{\substack{R_2 \rightarrow -1R_1 + R_2 \\ R_3 \rightarrow -2R_1 + R_3}} \begin{bmatrix} 2 & 4 & 6 \\ 0 & 1 & -3 \\ 0 & -9 & -10 \end{bmatrix} \xrightarrow{R_3 \rightarrow 9R_2 + R_3} \begin{bmatrix} 2 & 4 & 6 \\ 0 & 1 & -3 \\ 0 & 0 & -37 \end{bmatrix} \triangleq U$$

Check: No row exchange in the process; continue.

Step 2: Record elementary matrices.

$$E_1, E_2, E_3 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 9 & 1 \end{bmatrix}$$

Step 3: Compute L by (10.3).

$$L = E_1^{-1} E_2^{-1} E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 9 & 1 \end{bmatrix}^{-1}$$

opposite number

$$= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -9 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -9 & 1 \end{bmatrix}$$

Step 4: Conclusion.

A has an LU decomposition $A = LU$, where $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -9 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 4 & 6 \\ 0 & 1 & -3 \\ 0 & 0 & -37 \end{bmatrix}$

Claim of LU Decomposition: If No Row Exchange

Claim: Suppose A is a **square** matrix.

Suppose there is **no row exchange** in the process of GJ-E for solving $Ax=b$, where b is an arbitrary vector.

Then the matrix A can be decomposed as

$$A = LU,$$

where

- L is a **lower triangular matrix**;
- U is an **upper triangular matrix**.

What if there are row exchanges in GJ-E?

Not covered in this course.

[extended reading: PLU decomposition]

Exercise Question

Which of the following skills are necessary to compute an LU decomposition?

A) Understanding how to obtain E_i . ✓

B) Understanding how to obtain E_i^{-1} . ✓

C) Knowing how to multiply matrices. ✓

D) Knowing how to take the transpose of a matrix. X

E) Knowing how to add two matrices. X

[multiplying elementary matrices]
(not by definition,
but by row operations)

Reading Material
[NOT required in
quiz/homework/exam]
 $PA = LU$ for General A

Strang's book Sec 2.6

Main Result [just a result; no proof; no computation]

Theorem (PLU decomposition)

Any square matrix A can be written as

$$P A = LU,$$

where L is lower triangular,

U is upper triangular;

P is a certain permutation matrix.

PLU Decomposition in This Course:

--**Not required:** Manual computation of PLU; proof of the theorem.

--**Reason:** The proof is quite nontrivial; computation of PLU is a bit tricky.

Concluding Section



Summary Today

Write your summary below.

One sentence summary:

Detailed summary:

Summary Today Instructor's summary

One sentence summary:

We learned elementary matrices & LU decomposition.

Detailed summary:

- Three types of elementary matrices.
 - i) Obtained by apply the operation to I_n .
 - ii) Shape: $\bar{E}_{R_i R_j}$: not-triangular
 $\bar{E}_{R_i + R_j}$: upper or lower triangular
(depend on $i < j$ or $i > j$)
 $\bar{E}_{d R}$: diagonal
 - iii) inverse: same type of elementary matrix, same shape.
- GJ- $\bar{E}$ expressed as product of matrices: $\bar{E}_k \cdots \bar{E}_1 A = U$
- LU decomposition: $A = \bar{E}_1^{-1} \cdots \bar{E}_k^{-1} U$.